LMSR as Entropy-Smoothed Batch Auction Clearing

Proof Sketch

Draft — February 2026

Summary. We prove that Hanson's Logarithmic Market Scoring Rule (LMSR) is the entropy-smoothed version of a welfare-maximizing batch auction with minting. The connection is exact: the LP's minting cost function $V(D) = \max_k D_k$ and the LMSR cost function $C_b(D) = b \cdot \ln \sum_k \exp(D_k/b)$ are related by $C_b \to V$ as $b \to 0$. Their Fenchel conjugates exhibit the same convergence: the negative entropy $-b \cdot H(p)$ converges to the simplex indicator $\delta_\Delta(p)$. This establishes that LP batch clearing and LMSR pricing are endpoints of a single parametric family, and that the LP inherits all structural properties of LMSR (price normalization, arbitrage-freeness) as limiting cases.

1. The Minting Cost Function

1.1. Setup

Consider a prediction market with K mutually exclusive outcomes. In a batch auction, N orders arrive. After matching buyers against sellers, the residual imbalance is:

$$D_k = \sum_{i \in \text{ buy}(k)} q_i - \sum_{j \in \text{ sell}(k)} q_j \quad \text{for each outcome } k \in \{1, ..., K\}$$

where q_i is the fill quantity of order i. The vector $\boldsymbol{D} = (D_1, ..., D_K)$ is the *net demand* — the excess of buy volume over sell volume for each outcome.

To clear the market, the exchange must provide D_k additional shares of outcome k. Since outcomes are mutually exclusive, one *group mint* at cost \$1 creates one share of every outcome (exactly one resolves to \$1 at settlement, so this is fairly priced).

1.2. LP Minting Cost

The minimum minting cost to satisfy demand $\boldsymbol{D}$ is:

Definition 1 (Minting Cost Function).

$$V(\boldsymbol{D}) = \max_k D_k$$

Justification. Each group mint creates one share of every outcome. To supply D_k shares of outcome k for all k, we need at least $\max_k D_k$ mints (each mint covers one unit of every outcome). The cost is \$1 per mint, so the total cost is $\max_k D_k$. If $\max_k D_k < 0$ (all outcomes have excess supply), we *burn* pairs and recover revenue. ◇

The function $V = \max_k D_k$ is convex, piecewise linear, and non-smooth (it has kinks where $D_j = D_k$ for distinct j, k).

1.3. The Full Batch Auction as an Optimization

The welfare-maximizing batch clearing solves:

$$P: \quad \max_{\boldsymbol{q} \in [0, \overline{\boldsymbol{Q}}]} \quad \sum_i w_i q_i - V(\boldsymbol{D}(\boldsymbol{q}))$$

where w_i is the welfare coefficient of order i ($+L_i$ for buyers, $-L_i$ for sellers), and $\boldsymbol{D}(\boldsymbol{q})$ is the net demand vector (linear in $\boldsymbol{q}$). Since V is convex and $\boldsymbol{D}$ is linear in $\boldsymbol{q}$, the objective is concave — this is a convex optimization problem.

Introducing an explicit minting variable $\mu \geq \max_k D_k$, this is equivalent to the Linear Program:

$$\max_{\boldsymbol{q},\mu} \quad \sum_i w_i q_i - \mu \quad \text{s.t.} \quad D_k(\boldsymbol{q}) \leq \mu \quad \forall k, \quad \boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right]$$

2. Fenchel Duality: Prices from Conjugates

The structural properties of the LP clearing prices follow from the Fenchel conjugate of V.

2.1. The Conjugate of V

Theorem 1 (Minting–Simplex Duality). *The Fenchel conjugate of* $V(\boldsymbol{D}) = \max_k D_k$ *is the indicator function of the probability simplex:*

$$V^*(\boldsymbol{p}) = \sup_{\boldsymbol{D}} \left\{ \langle \boldsymbol{p}, \boldsymbol{D} \rangle - \max_k D_k \right\} = \delta_\Delta(\boldsymbol{p})$$

where $\Delta = \left\{ \boldsymbol{p} \in \mathbb{R}^K : p_k \geq 0, \sum_k p_k = 1 \right\}$ *and* $\delta_\Delta(\boldsymbol{p}) = 0$ *if* $\boldsymbol{p} \in \Delta$, $+\infty$ *otherwise.*

Proof. We compute $V^*(\boldsymbol{p}) = \sup_{\boldsymbol{D}} \left\{ \sum_k p_k D_k - \max_k D_k \right\}$ in three cases.

Case 1: $\boldsymbol{p} \in \Delta$. For any $\boldsymbol{D}$, we have $\sum_k p_k D_k \leq \max_k D_k$ (a convex combination of the D_k cannot exceed the maximum). So the supremum is ≤ 0. Setting $\boldsymbol{D} = \boldsymbol{0}$ gives value 0. Therefore $V^*(\boldsymbol{p}) = 0$.

Case 2: $\sum_k p_k > 1$. Take $\boldsymbol{D} = (t, t, ..., t)$ for $t \to +\infty$. Then $\sum p_k t - t = t(\sum p_k - 1) \to +\infty$.

Case 3: Some $p_j < 0$. Take $D_j \to -\infty$ with all other $D_k = 0$. Then $p_j D_j \to +\infty$ while $\max_k D_k = 0$, so the supremum diverges.

In Cases 2 and 3, $V^*(\boldsymbol{p}) = +\infty$. □

Interpretation. The batch auction's minting mechanism *is* the probability simplex. The constraint that prices must form a valid probability distribution is not imposed externally — it is the Fenchel conjugate of the minting cost. Minting at \$1 per complete set *encodes* the axiom $\sum p_k = 1$.

3. Log-Sum-Exp Smoothing

3.1. The LMSR Cost Function

Definition 2 (Smoothed Minting Cost). *For temperature parameter* $b > 0$*:*

$$C_b(\boldsymbol{D}) = b \cdot \ln \sum_{k=1}^{K} \exp\left(\frac{D_k}{b}\right)$$

This is the *log-sum-exp* (LSE) function, scaled by b. It is convex, smooth (C^∞), and approximates V:

Proposition 1 (LSE–Max Sandwich). *For all* $\boldsymbol{D} \in \mathbb{R}^K$*:*

$$\max_k D_k \leq C_b(\boldsymbol{D}) \leq \max_k D_k + b \ln K$$

In particular, $C_b(\boldsymbol{D}) \to V(\boldsymbol{D})$ *uniformly on compact sets as* $b \to 0^+$.

Proof. Lower bound: $\sum \exp(D_k/b) \geq \exp(\max_k D_k/b)$, so $C_b \geq \max_k D_k$. Upper bound: $\sum \exp(D_k/b) \leq K \exp(\max_k D_k/b)$, so $C_b \leq \max_k D_k + b \ln K$. □

The gap $b \ln K$ is precisely the maximum loss of the LMSR market maker — a well-known result that here falls out as an approximation bound.

3.2. The Conjugate of C_b

Theorem 2 (LMSR–Entropy Duality). *The Fenchel conjugate of C_b is the negative Shannon entropy, restricted to the simplex:*

$$C_b^*(\boldsymbol{p}) = \begin{cases} -b \cdot H(\boldsymbol{p}) = b \sum_k p_k \ln p_k & \text{if } \boldsymbol{p} \in \Delta \\ +\infty & \text{otherwise} \end{cases}$$

where $H(\boldsymbol{p}) = -\sum_k p_k \ln p_k$ is the Shannon entropy.

Proof. We compute $C_b^*(\boldsymbol{p}) = \sup_{\boldsymbol{D}} \{\langle \boldsymbol{p}, \boldsymbol{D} \rangle - b \ln \sum \exp(D_k/b)\}$.

Setting the gradient to zero:

$$\frac{\partial}{\partial D_k}\left[p_k D_k - b \ln \sum_j \exp\big(D_j/b\big)\right] = p_k - \frac{\exp(D_k/b)}{\sum_j \exp\big(D_j/b\big)} = 0$$

This gives $p_k = \exp(D_k/b)/\sum \exp\big(D_j/b\big)$ — the *softmax*, which is exactly the **LMSR marginal price**. Inverting: $D_k = b \ln p_k + b \ln Z$ where $Z = \sum \exp\big(D_j/b\big)$.

Substituting back:

$$\begin{aligned} \langle \boldsymbol{p}, \boldsymbol{D} \rangle &= \sum_k p_k (b \ln p_k + b \ln Z) = b \sum_k p_k \ln p_k + b \ln Z \\ C_b(\boldsymbol{D}) &= b \ln Z \end{aligned}$$

Therefore $C_b^*(\boldsymbol{p}) = b \sum_k p_k \ln p_k + b \ln Z - b \ln Z = b \sum_k p_k \ln p_k$.

This is finite only when $\boldsymbol{p} \in \Delta$ (the softmax always yields a probability vector). □

3.3. The Duality Diagram

The two pairs (V, V^*) and (C_b, C_b^*) form a commutative diagram under Fenchel conjugation and the $b \to 0$ limit:

$$\begin{array}{ccc} V(\boldsymbol{D}) = \max_k D_k & \xleftrightarrow{\text{Fenchel}} & V^*(\boldsymbol{p}) = \delta_\Delta(\boldsymbol{p}) \\ \uparrow\ b \to 0 & & \uparrow\ b \to 0 \\ C_b(\boldsymbol{D}) = b \ln \sum \exp(D_k/b) & \xleftrightarrow{\text{Fenchel}} & C_b^*(\boldsymbol{p}) = b \sum p_k \ln p_k \end{array}$$

Left column: **primal** (cost of minting shares). Right column: **dual** (constraint on prices).
Top row: **LP batch auction** (sharp). Bottom row: **LMSR** (smooth, temperature b).

As $b \to 0$: the smooth LMSR cost C_b sharpens to the LP minting cost V, and the soft entropy penalty $b \sum p_k \ln p_k$ hardens to the simplex indicator δ_Δ (a rigid constraint that prices must be probabilities).

4. LMSR Pricing as Smoothed Batch Clearing

4.1. The Smoothed Batch Auction

Replace the minting cost V with C_b in the batch clearing problem:

$$P_b : \max_{\boldsymbol{q} \in [0, \overline{\boldsymbol{Q}}]} \sum_i w_i q_i - C_b(\boldsymbol{D}(\boldsymbol{q}))$$

Since C_b is convex and smooth, and $\boldsymbol{D}(\boldsymbol{q})$ is linear, P_b is a smooth concave maximization. Its first-order conditions are necessary and sufficient.

Theorem 3 (Main Result). *At the optimum of P_b, the marginal cost of shares — i.e., the clearing prices — are:*

$$p_k^* = \frac{\partial C_b}{\partial D_k} = \frac{\exp(D_k^*/b)}{\sum_j \exp\left(D_j^*/b\right)}$$

This is the LMSR marginal price function.

Furthermore:
1. $\sum_k p_k^* = 1$ *(prices form a probability distribution).*
2. *As $b \to 0^+$, the solution of P_b converges to the solution of P (the LP batch auction).*
3. *As $b \to \infty$, prices converge to the uniform distribution $p_k = 1/K$.*

Proof of (1). The softmax always sums to 1 by construction.

Proof of (2). By Proposition 1, the objectives of P_b and P satisfy $|\mathrm{obj}(P_b) - \mathrm{obj}(P)| \leq b \ln K \to 0$. The feasible sets are identical ($\boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right]$). By standard results in epi-convergence of convex functions, the optimizers converge.

Proof of (3). As $b \to \infty$, $\exp(D_k/b) \to 1$ for all k, so $p_k \to 1/K$. □

4.2. Interpretation

The parameter b interpolates between two extremes:

- **$b = 0$ (LP batch auction):** Prices are sharp. The clearing price is set by the marginal order. Supply from minting is a step function: unlimited at \$1 per complete set, zero otherwise. Prices satisfy $\sum p_k = 1$ as a *hard* constraint (from $V^* = \delta_\Delta$).
- **$b > 0$ (LMSR):** Prices are smooth. The market maker provides a continuous supply curve with depth proportional to b. Prices satisfy $\sum p_k = 1$ by *softmax structure* (from $C_b^* = -bH$). The maximum loss from smoothing is $b \ln K$ (the entropy gap from Proposition 1).

The batch auction is the *zero-temperature* limit. Raising b adds liquidity (smoothness) at the cost of a bounded subsidy.

5. The KKT Conditions: Where the Exponentials Come From

To see exactly how LMSR pricing emerges from the optimality conditions, we write out the KKT system for P_b.

5.1. First-Order Conditions

For a buy order i on outcome k with welfare coefficient $w_i = L_i$:

$$\frac{\partial}{\partial q_i}[w_i q_i - C_b(\boldsymbol{D}(\boldsymbol{q}))] = L_i - \frac{\partial C_b}{\partial D_k} \cdot \frac{\partial D_k}{\partial q_i} = L_i - p_k$$

Combined with the box constraints $q_i \in \left[0, \overline{Q}_i\right]$ and multipliers μ_i^-, μ_i^+:

$$L_i - p_k - \mu_i^+ + \mu_i^- = 0, \quad \mu_i^+\left(q_i - \overline{Q}_i\right) = 0, \quad \mu_i^- q_i = 0$$

This gives the familiar clearing rule:

- $q_i = \overline{Q}_i$ (fully filled) $\Leftrightarrow$ $L_i \geq p_k$ *(buyer's limit above price)*
- $q_i = 0$ (unfilled) $\Leftrightarrow$ $L_i \leq p_k$ *(buyer's limit below price)*
- $0 < q_i < \overline{Q}_i$ (marginal) $\Leftrightarrow$ $L_i = p_k$ *(buyer is the price-setter)*

This is the *Uniform Clearing Price* (UCP) condition — identical to the LP case. The entropy smoothing does not alter the order-matching logic; it only changes how prices are determined from quantities.

5.2. Where the Exponentials Live

In the LP ($b = 0$), the price p_k is determined by the marginal order — the order whose limit price equals the clearing price. This is a *discrete* mechanism: the price jumps as the marginal order changes.

In P_b ($b > 0$), the price $p_k = \exp(D_k/b)/\sum \exp(D_j/b)$ depends *continuously* on the net demand D_k. The exponential arises from the first-order condition of the *minting cost*, not from the order book. The order book still determines fills via UCP; the exponential determines how the residual minting demand maps to prices.

6. Endogenous Liquidity

In Hanson's LMSR, the parameter b is chosen exogenously by the market creator. It represents a commitment to subsidize liquidity: the market maker accepts a bounded loss of $b \ln K$ to ensure smooth pricing. This is a *design choice*, not an emergent property.

> **Theorem 4** (Self-Financing Minting). *In the LP batch auction ($b = 0$), the minting mechanism is self-financing: at the optimal solution, the revenue from filled orders covers the minting cost exactly. The "market maker" (minting mechanism) incurs zero loss.*

Proof. The minting mechanism's profit-and-loss is: revenue from selling shares minus cost of minting.

$$\text{P\&L} = \underbrace{\sum_k p_k D_k}_{\text{revenue}} - \underbrace{\mu}_{\text{minting cost}}$$

where $\mu = \max_k D_k$ is the number of group mints and p_k is the clearing price of outcome k.

Recall the LP has constraints $D_k \leq \mu$ for each k, with dual variables $p_k \geq 0$. By complementary slackness: if $D_k < \mu$ (constraint not tight), then $p_k = 0$. Therefore $p_k > 0$ only for outcomes where $D_k = \mu$. This gives:

$$\sum_k p_k D_k = \sum_{k:D_k=\mu} p_k \cdot \mu = \mu \cdot \sum_{k:D_k=\mu} p_k$$

Since $p_k = 0$ for all k where $D_k < \mu$, and $\sum_k p_k = 1$ (Theorem 1), we have $\sum_{k:D_k=\mu} p_k = 1$. Therefore:

$$\text{P\&L} = \mu \cdot 1 - \mu = 0$$

The minting mechanism breaks even exactly. □

Contrast with LMSR. The smoothed cost $C_b > V$ (Proposition 1), so P_b collects *less* revenue than the LP for the same demand vector. The gap $C_b - V \leq b \ln K$ is exactly the LMSR subsidy. In the LP limit, this subsidy vanishes: liquidity is endogenous.

The economic intuition: LMSR provides smooth prices by *overpaying* for minting (the log-sum-exp exceeds the true max). The LP provides sharp prices by paying the exact cost. The order book — not an exogenous parameter — determines market depth.

7. Group Minting and Combinatorial Scaling

7.1. LMSR for Compound Securities

Hanson's combinatorial LMSR prices contracts over 2^K joint states when K markets are correlated. The cost function becomes:

$$C_b^{\text{comb}}(\boldsymbol{D}) = b \ln \sum_{s \in \{0,1\}^K} \exp(D_s / b)$$

where s ranges over all 2^K joint outcomes. Each evaluation of the cost function is $O(2^K)$, and the pricing requires the full state vector.

For $K = 20$ (a medium election), this is $2^{20} \approx 10^6$ states. For $K = 50$ (a large election), it is computationally intractable.

7.2. Group Minting: $O(K)$ Instead of $O(2^K)$

For mutually exclusive outcomes (exactly one of K outcomes realizes), the LP's group minting cost is:

$$V_G(D_1, ..., D_K) = \max_{k=1}^{K} D_k$$

This is *identical* to the binary case (Definition 1) but over K terms instead of 2. The smoothed version:

$$C_b^G(D_1, ..., D_K) = b \ln \sum_{k=1}^{K} \exp(D_k / b)$$

This is K-outcome LMSR — computed in $O(K)$ time, not $O(2^K)$.

> **Proposition 2** (Scaling Advantage). *For a group of K mutually exclusive markets, the LP batch auction requires $O(K)$ balance constraints and one group minting variable. The equivalent combinatorial LMSR requires $O(2^K)$ state evaluations. The LP achieves the same price normalization ($\sum_k p_k^{\text{YES}} \leq 1$) and the same limiting behavior ($C_b^G \to V_G$ as $b \to 0$), with an exponential reduction in complexity.*

The structural insight: mutual exclusivity means the 2^K-dimensional joint state space collapses to K states (exactly one outcome is YES). Group minting exploits this directly. Combinatorial LMSR must discover it through the cost function.

8. Budget-Constrained Clearing via Frank-Wolfe

The entropy framework developed in §§1–7 handles the LP core exactly. We now address the remaining non-convexity: the bilinear market maker budget constraint.

8.1. Problem Setup

A market maker k deposits balance B_k and posts orders across multiple markets. The capital consumed by each fill depends on the clearing price:

$$\text{cap}_k(\boldsymbol{p}, \boldsymbol{q}) = \sum_{i \in \text{MM}_k} c_i\left(p_{m(i)}\right) \cdot q_i, \quad c_i(p) = \begin{cases} p & \text{if BuyYes/SellNo} \\ 1 - p & \text{if SellYes/BuyNo} \end{cases}$$

The budget constraint $\text{cap}_k \leq B_k$ is bilinear: p is determined by $\boldsymbol{q}$ through the clearing mechanism (LP duality or LMSR softmax). We seek:

$$\max_{\boldsymbol{q} \in \mathcal{C}} \quad f(\boldsymbol{q}) \quad \text{s.t.} \quad \text{cap}_k(\boldsymbol{p}(\boldsymbol{q}), \boldsymbol{q}) \leq B_k \quad \forall k$$

where $\mathcal{C}$ is the LP feasible set (balance constraints, box bounds, minting) and f is welfare minus minting cost.

8.2. Lagrangian Relaxation of Budgets

Dualize the K budget constraints with multipliers $\mu_k \geq 0$:

$$\mathcal{L}(\boldsymbol{q}, \boldsymbol{\mu}) = f(\boldsymbol{q}) + \sum_k \mu_k (B_k - \mathrm{cap}_k(\boldsymbol{q}))$$

For fixed $\boldsymbol{\mu}$, maximizing $\mathcal{L}$ over $\boldsymbol{q} \in \mathcal{C}$ decouples the budget from the LP structure. The budget enters only through *modified welfare coefficients*:

$$w_i' = w_i - \sum_{k : i \in \mathrm{MM}_k} \mu_k \cdot c_i(p^t)$$

where p^t is the current price estimate. The Lagrangian subproblem $\max_{\boldsymbol{q} \in \mathcal{C}} \sum w_i' q_i$ is a standard LP — the same LP as our base solver, with adjusted objectives. This is the key structural insight: **the budget shadow prices modify each MM order's effective welfare, and the LP handles everything else**.

8.3. The Algorithm

Algorithm 1: Frank-Wolfe with Lagrangian Budget Handling

Input: Orders, markets, groups, MM budgets $B_1, ..., B_K$. Temperature $b > 0$.

Initialize: Solve base LP (no budgets) $\to \boldsymbol{q}^0$, $\boldsymbol{p}^0$. Set $\boldsymbol{\mu} = \boldsymbol{0}$.

For $t = 0, 1, 2, ...$:

1. **Prices.** Compute $p_m^t = \mathrm{softmax}(D_m(\boldsymbol{q}^t)/b)$ for each market m.
2. **Budget evaluation.** For each MM k: $\mathrm{cap}_k^t = \sum_{i \in \mathrm{MM}_k} c_i(p^t) \cdot q_i^t$.
3. **Modified welfare.** $w_i' = w_i - \sum_k \mu_k \cdot c_i(p^t)$ for each order i.
4. **Frank-Wolfe oracle.** Solve LP: $\boldsymbol{s} = \mathrm{argmax}_{\boldsymbol{q} \in \mathcal{C}} \sum w_i' q_i$.
5. **Step.** $\boldsymbol{q}^{t+1} = (1 - \gamma_t)\boldsymbol{q}^t + \gamma_t \boldsymbol{s}, \quad \gamma_t = \frac{2}{t+2}$.
6. **Dual update.** $\mu_k \leftarrow \max(0, \mu_k + \eta_t(\mathrm{cap}_k^t - B_k))$ for each MM k.

Terminate when $\max_k (\mathrm{cap}_k - B_k)^+ < \varepsilon$. Round final $\boldsymbol{q}$ to nearest LP vertex.

Each iteration requires: one softmax evaluation ($O(M)$), one LP solve (< 1ms), and one dual update ($O(K)$). The LP in step 4 is structurally identical to the base clearing LP — the budget information enters solely through the modified welfare coefficients w_i'.

8.4. Why This Is Not SLP

Our current Sequential LP corresponds to Algorithm 1 with $\gamma = 1$ (jump to the LP solution) and $\mu = 0$ (no dual variable, budget handled as a hard constraint via linearization). This has two failure modes:

1. **Over-stepping ($\boldsymbol{\gamma = 1}$):** Jumping to the LP vertex can overshoot. The new prices at the vertex may violate the budget constraint despite the linearized constraint being satisfied (§6 counterexample).

2. **No dual memory ($\boldsymbol{\mu = 0}$):** Each SLP iteration starts fresh. Information about budget tightness from previous iterations is discarded.

Frank-Wolfe fixes both: the step size $\gamma_t = 2/(t + 2)$ ensures gradual convergence, and the dual variable μ_k accumulates budget shadow price information across iterations.

8.5. Convergence

Theorem 7 (Frank-Wolfe Convergence). *For fixed $b > 0$, Algorithm 1 with step size $\gamma_t = 2/(t+2)$ and dual step size $\eta_t = 1/\sqrt{t}$ satisfies:*

$$\mathcal{L}(\boldsymbol{q}^*, \boldsymbol{\mu}^*) - \mathcal{L}(\boldsymbol{q}^t, \boldsymbol{\mu}^t) \leq O\left(L_b \cdot \frac{D^2}{t} + \frac{B_{\max}}{\sqrt{t}}\right)$$

where L_b is the Lipschitz constant of ∇f_b (proportional to $1/b$), $D = \operatorname{diam}(\mathcal{C})$, and $B_{\max} = \max_k B_k$.

Proof sketch. The primal update (step 5) is standard Frank-Wolfe over a compact convex set, giving $O(L_b D^2/t)$ convergence of the primal objective for fixed $\boldsymbol{\mu}$. The dual update (step 6) is projected subgradient ascent on the concave dual function, giving $O\left(1/\sqrt{t}\right)$ convergence of the dual. The combined primal-dual rate is dominated by the slower dual convergence. □

Remark. The Lipschitz constant $L_b \propto 1/b$ means that sharper pricing (smaller b) requires more iterations. This motivates **annealing**: start with large b (fast convergence, smooth prices), decrease b as the algorithm progresses (sharper prices, warm-started from previous solution).

8.6. Annealing Schedule

Algorithm 2: Annealed Frank-Wolfe

Set b_0 (e.g., $0.1 \cdot \$1$), $b_{\min}$ (e.g., $10^{-4} \cdot \$1$), cooling factor $\rho = 0.5$.

For $\ell = 0, 1, 2, \ldots$ while $b_\ell > b_{\min}$:

  Run Algorithm 1 for T iterations at temperature b_ℓ.

  Warm-start next round: $\boldsymbol{q}^0 \leftarrow \boldsymbol{q}^T$, $\boldsymbol{\mu}^0 \leftarrow \boldsymbol{\mu}^T$.

  Cool: $b_{\ell+1} = \rho \cdot b_\ell$.

Final: solve LP at $\boldsymbol{q}^{\text{final}}$ to obtain integer fills and exact dual prices.

Total cost: $\left\lceil \log_{1/\rho}(b_0/b_{\min}) \right\rceil \times T$ LP solves. For $b_0 = 0.1$, $b_{\min} = 10^{-4}$, $\rho = 0.5$, $T = 5$: approximately $10 \times 5 = 50$ LP solves, or $\sim$ 50ms.

8.7. What This Proves and What It Doesn't

Proven:

- Algorithm 1 converges to a KKT point of the smoothed budget-constrained problem at rate $O\left(1/\sqrt{t}\right)$.
- The final LP solve (Algorithm 2, last step) produces exact integer fills with correct dual prices.
- Each iteration reuses the existing LP infrastructure with zero additional solver dependencies.

Not proven:

- Global optimality. The bilinear constraint makes the feasible set non-convex. The KKT point found by Frank-Wolfe may be a local optimum. However, empirically, the LP-optimal prices (step 0) are close to the budget-constrained optimum — the budget perturbation is small — so the Frank-Wolfe trajectory stays in the basin of the global optimum.
- Formal convergence rate as $b \to 0$. The annealing schedule is heuristic. A rigorous analysis would require bounding the path-following error across temperature steps (similar to interior point path-following theory).

What would close the gap: A proof that the budget-constrained problem has a unique KKT point for generic instances. We attempt this below.

8.8. Global Optimality via Budget Slippage

We now prove that under a natural economic condition, the budget-constrained problem has a unique KKT point — making Frank-Wolfe globally optimal. The key insight: **price slippage makes the budget constraint convex**, which preserves the concavity of the Lagrangian.

8.8.1. The Lagrangian

$$\mathcal{L}(\boldsymbol{q}, \boldsymbol{\mu}) = \underbrace{f_b(\boldsymbol{q})}_{\text{concave}} - \sum_k \mu_k \cdot \underbrace{\mathrm{cap}_k(\boldsymbol{q})}_{\text{convex?}} + \sum_k \mu_k B_k$$

The objective f_b is concave (linear minus convex). If each cap_k is convex in $\boldsymbol{q}$, then $-\mu_k \cdot \mathrm{cap}_k$ is concave for $\mu_k \geq 0$. The full Lagrangian would then be strictly concave in the price-relevant subspace, giving a unique primal maximizer $\boldsymbol{q}^*(\boldsymbol{\mu})$ for each $\boldsymbol{\mu}$, a convex dual function $d(\boldsymbol{\mu}) = \max_{\boldsymbol{q}} \mathcal{L}$, and a unique saddle point $(\boldsymbol{q}^*, \boldsymbol{\mu}^*)$ — i.e., a unique KKT point.

Everything hinges on: **is the capital function convex?**

8.8.2. The Hessian of price × quantity

Consider a single BuyYes order i on binary market m with aggregate MM fill Q on that market. The capital contribution is $\mathrm{cap} = p_m(Q) \cdot Q$ where $p_m = \sigma((Q + R)/b)$ is the sigmoid (softmax for $K = 2$), with R capturing all non-MM demand (retail orders, other-side fills) as a constant.

Computing the second derivative:

$$\frac{d^2}{dQ^2}[p \cdot Q] = \frac{p(1-p)}{b} \cdot \left[2 + \frac{Q(1-2p)}{b}\right]$$

The prefactor $p(1-p)/b > 0$ always. The sign depends on the bracket.

Proposition 3 (Budget Slippage Convexity). *The capital function* $\mathrm{cap}(Q) = p(Q) \cdot Q$ *is convex in* Q *if and only if:*

$$Q \cdot (2p - 1) \leq 2b$$

For $p \leq 1/2$ (buying the underdog), this holds unconditionally. For $p > 1/2$ (buying the favorite), this requires $Q \leq 2b/(2p-1)$.

Economic meaning. Convexity of capital = price slippage acts as a brake. Buying more shares pushes the price up, making each additional share *more* expensive. This self-correcting feedback is what prevents multiple equilibria.

The condition fails when $Q \gg b$: the sigmoid saturates ($p \to 1$), slippage vanishes (the price can't go higher), and the brake disengages. Economically: an MM buying so aggressively that the price is already near \$1 faces no further slippage penalty — the feedback loop is broken.

8.8.3. The group Hessian

For mutually exclusive outcomes in a group, the softmax couples all prices: increasing Q_A raises p_A but lowers p_B (the denominator grows). Rather than bounding individual cross-terms, we compute the full quadratic form directly.

Proposition 4 (Group Budget Slippage). *Let $\boldsymbol{Q} = (Q_1, ..., Q_K)$ be MM k's fill vector on a group of K mutually exclusive outcomes with softmax prices $\boldsymbol{p} = \mathrm{softmax}((\boldsymbol{Q} + \boldsymbol{R})/b)$. The Hessian of the total capital $\mathrm{cap}(\boldsymbol{Q}) = \sum_k p_k Q_k$ satisfies:*

$$\boldsymbol{v}^\top \nabla^2 \ \mathrm{cap} \cdot \boldsymbol{v} = \frac{1}{b} \sum_{k=1}^{K} p_k (v_k - \overline{v})^2 \cdot \left(2 + \left(Q_k - \overline{Q}\right)/b\right)$$

where $\overline{v} = \sum_k p_k v_k$ and $\overline{Q} = \sum_k p_k Q_k$.

The Hessian is positive semidefinite if and only if:

$$\overline{Q} - Q_k \leq 2b \quad \forall k$$

Proof. The capital function is $g(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{p}(\boldsymbol{x})$ where $\boldsymbol{p} = \mathrm{softmax}((\boldsymbol{x} + \boldsymbol{R})/b)$. Its gradient is $\nabla g_i = p_i \alpha_i$ where $\alpha_i = 1 + (x_i - \overline{x})/b$ and $\overline{x} = \sum_k p_k x_k$. (This follows from the product rule and the softmax Jacobian $J = (1/b)(\mathrm{diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\top)$.)

For the second directional derivative along $\boldsymbol{v}$, we compute $d^2 g(\boldsymbol{x} + t\boldsymbol{v})/dt^2 \ |_{t=0}$. Using $\dot{p}_i = (p_i/b)(v_i - \overline{v})$ and $\dot{\alpha}_i = (v_i - \overline{v})/b - \mathrm{Cov}_p(v, x)/b^2$, the cross-terms involving $\mathrm{Cov}_p(v, x)$ cancel exactly, leaving:

$$\boldsymbol{v}^\top H \boldsymbol{v} = \frac{1}{b} \sum_k p_k (v_k - \overline{v})^2 (2 + (x_k - \overline{x})/b)$$

Since each term has weight $p_k (v_k - \overline{v})^2 / b \geq 0$, the sign is determined by the bracket $2 + (x_k - \overline{x})/b$. The quadratic form is non-negative for all $\boldsymbol{v}$ iff this bracket is non-negative for all k, giving $\overline{x} - x_k \leq 2b$. Substituting $x_k = Q_k$ yields the stated condition. □

Verification. Setting $K = 2$, $Q_2 = 0$, $\boldsymbol{v} = (1, 0)$: the formula reduces to $(p(1-p)/b)[2 + Q(1-2p)/b]$, recovering Proposition 3 exactly.

8.8.4. Interpretation: capital bound per group

For the common case where an MM has orders on a strict subset $S \subsetneq \{1, ..., K\}$ of outcomes (e.g., BuyYes on a few favorites), the binding condition comes from outcomes $k \notin S$ where $Q_k = 0$:

$$\overline{Q} - 0 = \overline{Q} \leq 2b$$

Since $\overline{Q} = \sum_{k \in S} p_k Q_k$ is precisely the MM's total capital on the group, this simplifies to:

$$\textbf{Group DIC}: \quad \mathrm{cap}^k_{\mathrm{group}} \leq 2b$$

The MM's capital deployed on any single group must be at most $2b$. This is a natural liquidity condition: the entropy temperature must be at least half the capital exposure per group. For MMs practicing flash liquidity (spreading capital across many groups), this is easily satisfied.

8.8.5. The multi-group Hessian

For MM k with fills across groups $g_1, ..., g_G$ belonging to *different* group hierarchies (independent groups), the Hessian of cap_k is block-diagonal — one block per group. The full Hessian is PSD iff each block is PSD (Proposition 4 applied per group).

8.8.6. The Diluted Influence Condition

Definition 3 (Diluted Influence). *An instance satisfies the Diluted Influence Condition at temperature b if:*

1. *For every MM k and every independent (non-grouped) binary market m:* $Q^k_m \cdot (2p_m - 1)^+ \leq 2b$

2. *For every MM k and every group g:* $\quad \overline{Q}_g^k = \sum_{m \in g} p_m Q_m^k \leq 2b$

where Q_m^k is MM k's total fill on market m and p_m is the clearing price.

Condition (1) ensures convexity of capital per independent market (Proposition 3). Condition (2) ensures convexity of capital per group (Proposition 4). Condition (2) is stricter: it requires the MM's total capital on the group to be at most $2b$, whereas (1) only requires the "directional" capital $Q(2p-1)$ per market to be bounded.

Both conditions hold naturally when:
- b is large (high temperature, smooth prices — automatic for the first phase of annealing)
- MM fills are small relative to total volume (retail dilution)
- MMs are buying underdogs ($p < 0.5$, condition (1) is automatic)
- MMs spread capital across many groups (flash liquidity — capital per group stays small)

8.8.7. Uniqueness theorem

Theorem 8 (Global Optimality under Diluted Influence). *If the Diluted Influence Condition (Definition 3) holds at temperature b, then:*

1. *The capital function* cap_k *is convex in* $\boldsymbol{q}$ *for each MM* k *— including on grouped markets.*
2. *The Lagrangian* $\mathcal{L}(\boldsymbol{q}, \boldsymbol{\mu})$ *is strictly concave in* $\boldsymbol{q}$ *on the price-relevant subspace for any* $\boldsymbol{\mu} \geq 0$.
3. *The KKT point* $(\boldsymbol{q}^*, \boldsymbol{\mu}^*)$ *is unique (up to demand-preserving fill substitutions).*
4. *Algorithm 1 converges to the global optimum.*

Proof. (1): For independent markets, Proposition 3 and DIC condition (1). For groups, Proposition 4 and DIC condition (2). Since the Hessian of cap_k is block-diagonal across independent groups, PSD of each block gives PSD of the whole. (2): $\mathcal{L} = f_b - \sum \mu_k \mathrm{cap}_k + \mathrm{const}$. The term f_b is strictly concave in the price-relevant subspace. Each $-\mu_k \mathrm{cap}_k$ is concave (for $\mu_k \geq 0$ and convex cap_k). Their sum is strictly concave. (3): For fixed $\boldsymbol{\mu}$, strict concavity gives a unique maximizer $\boldsymbol{q}^*(\boldsymbol{\mu})$. The dual function $d(\boldsymbol{\mu}) = \mathcal{L}(\boldsymbol{q}^*(\boldsymbol{\mu}), \boldsymbol{\mu})$ is convex (supremum of affine functions of $\boldsymbol{\mu}$). For generic parameters, d is strictly convex at its minimum, giving a unique $\boldsymbol{\mu}^*$. (4): Frank-Wolfe on a concave Lagrangian converges to the unique saddle point. □

8.8.8. Quantitative contraction threshold

The softmax operator is exactly $1/(2b)$-Lipschitz continuous across all L_p norms — a tight bound, improving the commonly assumed $1/b$. This gives a precise contraction threshold: the best-response Jacobian (mapping current fills to LP-optimal fills via price adjustment) has spectral radius < 1 when:

$$\frac{1}{2b} \cdot \|A\|_\infty < 1, \quad \text{i.e.,} \quad b > \|A\frac{\|_\infty}{2}$$

where A is the demand matrix ($D = A\boldsymbol{q}$) and $\|A\|_\infty$ is its max row sum. This is the **quantitative** version of the Diluted Influence Condition: when b exceeds this threshold, the price-fill feedback is a global contraction — guaranteeing unique equilibrium and exponential convergence.

For the annealing schedule (Algorithm 2), this means: set $b_0 \geq \|A\frac{\|_\infty}{2}$ to ensure the first phase provably finds the global optimum.

8.8.9. Why unconditional uniqueness fails

Unconditional uniqueness — for *all* parameters and all b — is false. The entropy curvature (from C_b) and the budget non-convexity (from cap_k) both scale as $O(1/b)$. Neither dominates the other in general, so the budget can create a “ridge” in the Lagrangian landscape that splits the global optimum into two local optima.

Counterexample. Two markets A, B with identical order books and one MM with symmetric positions on both. By symmetry, there are two KKT points: “fill A, starve B” and “fill B, starve A.” These produce different fill vectors, different demands ($D_A > D_B$ vs. $D_B > D_A$), and different prices ($p_A > p_B$ vs. $p_B > p_A$). No algorithm can select between them without breaking the symmetry.

This counterexample requires *exact* parameter symmetry — a measure-zero set. Any perturbation of a single limit price selects a unique optimum.

8.8.10. Generic uniqueness

The counterexample is essentially the only obstruction. For almost all parameter values, the budget-constrained problem has a unique KKT point.

> **Theorem 9** (Generic Uniqueness). *Fix $b > 0$. Let Θ denote the space of problem parameters (limit prices, budgets, max fills). The set of parameters $\theta \in \Theta$ for which $P_b(\theta)$ has multiple KKT points is closed and has Lebesgue measure zero.*

Proof. The KKT system for the smoothed problem P_b is C^∞ (softmax is smooth for $b > 0$). For each active set — which orders have interior fills, which budgets bind — the KKT conditions form a square system $F(z, \theta) = 0$ where $z = \left(q_I, \mu_{K_a}\right)$.

Step 1 (Non-degeneracy). The derivative $D_\theta F$ has full rank: perturbing limit price w_i shifts the i-th stationarity equation independently; perturbing budget B_k shifts the k-th binding equation. By the Parametric Transversality Theorem (Abraham & Robbin, 1967; a consequence of Sard’s theorem), for almost all θ, 0 is a regular value of $F(\cdot, \theta)$. At regular values, the KKT Jacobian $D_z F$ is nonsingular at every solution: each KKT point is non-degenerate. Finitely many active set choices $\to$ union of exceptional parameters still measure zero. Compactness of $\mathcal{C} \to$ finitely many KKT points.

Step 2 (Homotopy from large b). At $b_0 > \|A\|_\infty /2$, the contraction bound gives exactly one KKT point. Consider the homotopy $b : b_0 \to b_{\min}$. Non-degenerate KKT points trace smooth paths in (z, b)-space (implicit function theorem). New KKT points appear only via *saddle-node bifurcation*: a max-saddle pair is born (or annihilated) when the bordered Hessian becomes singular. Bifurcations are codimension-1 in (b, θ)-space; for generic θ, they are isolated in b.

Step 3 (Global max persistence). The global optimum value $V(b) = \max f_b(\boldsymbol{q})$ over the feasible set is continuous in b (Berge’s Maximum Theorem). At a bifurcation, the newborn local maximum has welfare *equal to its twin saddle* — generically strictly below $V(b)$. The global maximum therefore continues as the smooth deformation of the unique optimum at b_0, without jumping or splitting. □

Corollary. For generic parameters, Algorithm 2 (Annealed Frank-Wolfe) with $b_0 \geq \|A\|_\infty /2$ converges to the global optimum at every temperature along the annealing path.

Economic meaning. The only instances with multiple optima require exact symmetries: identical order books, symmetric MM positions, matching budgets. Real order books — with heterogeneous beliefs, diverse limit prices, and asymmetric market structures — are generically unique. Theorem 8 (DIC) provides a *checkable certificate* for specific instances; Theorem 9 says the certificate is almost never needed.

8.8.11. Demand diameter bound

Even when multiple KKT points exist (the measure-zero case), their demands and prices cannot be far apart.

Proposition 5 (Demand Diameter). *Let $\boldsymbol{q}^1, \boldsymbol{q}^2$ both maximize $\mathcal{L}(\cdot, \boldsymbol{\mu})$ over $\mathcal{C}$ for fixed $\boldsymbol{\mu} \geq 0$ and $b > 0$. Then:*

$$\|\boldsymbol{D}^1 - \boldsymbol{D}^2\|_2 \leq \frac{b \cdot \sum_k \mu_k \overline{Q}_k}{p_{\min}}$$

where $\boldsymbol{D}^j = A\boldsymbol{q}^j$, $\overline{Q}_k = \max_j \sum_{i \in \mathrm{MM}_k} q_i^j$, and $p_{\min} = \min_m p_m(b)$. In particular, $\boldsymbol{\mu} = 0$ implies $\boldsymbol{D}^1 = \boldsymbol{D}^2$: demands and prices are unique when no budget binds.

Proof. The midpoint $\overline{\boldsymbol{q}} = (\boldsymbol{q}^1 + \boldsymbol{q}^2)/2$ lies in $\mathcal{C}$ (convex set). Its Lagrangian value:

$$\mathcal{L}(\overline{\boldsymbol{q}}, \boldsymbol{\mu}) - \frac{1}{2}(\mathcal{L}(\boldsymbol{q}^1, \boldsymbol{\mu}) + \mathcal{L}(\boldsymbol{q}^2, \boldsymbol{\mu})) = \underbrace{\delta}_{\text{entropy gain}} - \underbrace{\varepsilon}_{\text{cap perturbation}}$$

The entropy gain: $\delta = 1/2(C_b(\boldsymbol{D}^1) + C_b(\boldsymbol{D}^2)) - C_b(\overline{\boldsymbol{D}}) \geq \frac{p_{\min}}{2b} \|\boldsymbol{D}^1 - \boldsymbol{D}^2\|_2^2$ by strict convexity of C_b (minimum Hessian eigenvalue $= p_{\min}/b$).

The cap perturbation: $|\varepsilon| = |\sum_k \mu_k[\mathrm{cap}_k(\overline{\boldsymbol{q}}) - 1/2(\mathrm{cap}_k(\boldsymbol{q}^1) + \mathrm{cap}_k(\boldsymbol{q}^2))]|$. Each $|\mathrm{cap}_k(\overline{\boldsymbol{q}}) - 1/2(...)| \leq \frac{\overline{Q}_k}{4b} \|\boldsymbol{D}^1 - \boldsymbol{D}^2\|_2$ by the $1/(2b)$-Lipschitz bound on softmax prices.

Since $\overline{\boldsymbol{q}}$ cannot exceed the optimal value: $\delta \leq |\varepsilon|$. Dividing through:

$$\frac{p_{\min}}{2b} \|\boldsymbol{D}^1 - \boldsymbol{D}^2\|_2 \leq \frac{\sum_k \mu_k \overline{Q}_k}{4b}$$

Rearranging gives the bound. When $\boldsymbol{\mu} = 0$, the RHS vanishes, forcing $\boldsymbol{D}^1 = \boldsymbol{D}^2$. □

Remark. Proposition 5 interpolates between two regimes. When budgets are slack ($\boldsymbol{\mu} = 0$), the strict concavity of C_b dominates and demands are unique — the problem is "essentially convex." As budgets become tighter (larger μ_k), the cap perturbation grows, allowing wider demand separation. The DIC (Theorem 8) is the condition under which the perturbation vanishes entirely (convex cap → no perturbation → unique demand regardless of μ).

8.8.12. The Fenchel dual and price uniqueness

A natural hope: even if *fills* are non-unique, perhaps *clearing prices* are unconditionally unique. We test this via the Fenchel dual.

Unconstrained case (no budgets). The primal in demand space is $\max_D[W(D) - C_b(D)]$ where $W(D) = \max_{\boldsymbol{q}: A\boldsymbol{q}=D, \boldsymbol{q} \in \text{ box}} \sum w_i q_i$ is concave piecewise-linear. Define $h = -W$ (convex). By Fenchel-Rockafellar duality:

$$\min_{\boldsymbol{p} \in \Delta^M} \left[\underbrace{W^*(\boldsymbol{p})}_{\text{consumer surplus}} + \underbrace{C_b^*(\boldsymbol{p})}_{\text{entropy penalty}} \right]$$

where $W^*(\boldsymbol{p}) = \sum_i \left(w_i - (A^\top \boldsymbol{p})_i\right)^+ \overline{q}_i$ is the total surplus of orders whose limit prices exceed clearing prices, and $C_b^*(\boldsymbol{p}) = \sum_m b \sum_k p_{mk} \ln p_{mk}$ is the negative Shannon entropy (Theorem 2).

Proposition 6 (Unconstrained Price Uniqueness). *The Fenchel dual of the unconstrained smoothed problem is strictly convex: W^* is convex (piecewise linear) and C_b^* is strictly convex on the interior of Δ^M. The clearing prices $\boldsymbol{p}^*$ are therefore unique for any $b > 0$ and any order book, without any condition.*

This is the Fenchel dual of Theorems 1–2: the indicator function δ_Δ (hard simplex constraint) generalizes to the entropy penalty $-bH(\boldsymbol{p})$ (soft simplex constraint). The entropy's strict convexity is what makes prices unique — it penalizes ambiguity.

Budget-constrained case. Replace $W(D)$ with $W_B(D) = \max_{\boldsymbol{q}\in\mathcal{C},\mathrm{cap}_k\le B_k} \sum w_i q_i$ subject to $A\boldsymbol{q} = D$. The cap constraint $\sum_{i\in\mathrm{MM}_k} c_i(\mathrm{softmax}(D/b))q_i \le B_k$ depends on D through the softmax prices, making W_B *non-concave* in D.

The Fenchel dual becomes $\min_{\boldsymbol{p}}[W_B^*(\boldsymbol{p}) + C_b^*(\boldsymbol{p})]$, but now W_B^* is *non-convex.* The non-convexity arises from the cap constraint's dependence on D: as D shifts, the softmax prices change at rate $O(1/b)$, modifying which fills are budget-feasible. This introduces curvature of order $O(1/b)$ into W_B — the same scale as the entropy Hessian $\nabla^2 C_b^* = \mathrm{diag}(1/\boldsymbol{p})/b$.

Proposition 7 (Cross-Price Obstruction). *Unconditional price uniqueness is impossible for the budget-constrained smoothed problem. Specifically, the standard monotonicity argument fails due to cross-price budget violation.*

Proof. Suppose two KKT points $(\boldsymbol{q}^1, \boldsymbol{p}^1, \boldsymbol{\mu}^1)$ and $(\boldsymbol{q}^2, \boldsymbol{p}^2, \boldsymbol{\mu}^2)$ exist with $\boldsymbol{p}^1 \neq \boldsymbol{p}^2$. The strict convexity of C_b^* gives $\langle \boldsymbol{D}^1 - \boldsymbol{D}^2, \boldsymbol{p}^1 - \boldsymbol{p}^2\rangle > 0$ (the demand-price inner product is strictly positive for distinct prices). Cross-applying the optimality conditions and using complementary slackness ($\mu_k^j(B_k - E_k(\boldsymbol{q}^j, \boldsymbol{p}^j)) = 0$) yields:

$$\langle \boldsymbol{D}^1 - \boldsymbol{D}^2, \boldsymbol{p}^1 - \boldsymbol{p}^2\rangle \le \sum_k [\mu_k^2(E_k(\boldsymbol{q}^1, \boldsymbol{p}^2) - B_k) + \mu_k^1(E_k(\boldsymbol{q}^2, \boldsymbol{p}^1) - B_k)]$$

For a contradiction (LHS > 0 but RHS ≤ 0), we need $E_k(\boldsymbol{q}^1, \boldsymbol{p}^2) \le B_k$ — that the *first* fills are affordable at the *second* prices. But while $E_k(\boldsymbol{q}^1, \boldsymbol{p}^1) \le B_k$ holds (budget feasibility at own prices), there is **no guarantee** that $E_k(\boldsymbol{q}^1, \boldsymbol{p}^2) \le B_k$. If $\boldsymbol{p}^2$ is higher in markets where MM k holds long positions, the cross-price evaluation blows past the budget.

This is the mathematical signature of a *Generalized Nash Equilibrium Problem* (GNEP): the feasible set (budget constraint) depends on the dual variable (prices). Standard Fenchel duality requires the primal feasible set to be independent of the dual; the endogenous-price budget destroys this independence. The obstruction is structural, not a gap in technique: both the entropy curvature and the cross-price budget violation scale as $O(1/b)$, preventing either from dominating unconditionally. □

Remark. This is the dual-space version of the symmetric counterexample. In the primal, two optima have different fills and different prices. In the dual, the budget feasibility "swaps" — fills affordable at one price are not affordable at the other — because the bilinear constraint $c(p) \cdot q \le B$ ties the feasible region to the solution.

8.8.13. The expenditure perspective

Why Eisenberg-Gale fails for linear welfare. Changing variables to capital expenditures $e_i = c_i(p) \cdot q_i$ linearizes the budget: $\sum_{i\in\mathrm{MM}_k} e_i \le B_k$. But welfare transforms to $\sum_i w_i \cdot e_i / c_i(p)$ — rational in prices. In Fisher markets (Eisenberg & Gale, 1959), the analogous program is convex because agents have *diminishing-returns* utilities: $\sum B_k \ln U_k$ provides curvature. Our MMs have constant marginal returns (linear welfare), providing none.

8.8.14. Risk-averse batch clearing

The expenditure substitution *does* work when market makers have diminishing returns. Replace linear MM welfare with Kelly-criterion utility: each additional fill has diminishing marginal value, reflecting the risk aversion that real institutional MMs exhibit.

Theorem 10 (Risk-Averse Clearing is Convex). *Define the risk-averse batch auction clearing program:*

$$P_b^{\mathrm{RA}}: \quad \max_{\boldsymbol{q}\in\mathcal{C}} \quad \underbrace{\sum_k B_k \ln U_k(\boldsymbol{q})}_{\text{MM welfare}} + \underbrace{\sum_{j\notin \text{ MM}} w_j q_j}_{\text{retail welfare}} - \underbrace{C_b(\boldsymbol{D}(\boldsymbol{q}))}_{\text{minting cost}}$$

where $U_k(\boldsymbol{q}) = \sum_{i\in \mathrm{MM}_k} w_i q_i$ *is MM k's total weighted fill,* $\mathcal{C} = \left\{\boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right] : \text{balance constraints}\right\}$ *is the LP feasible set, and* C_b *is the smoothed minting cost (Definition 2). Then:*

1. *The objective is strictly concave on the feasible set (whenever any MM has positive-welfare orders with* $B_k > 0$*).*
2. P_b^{RA} *has a unique optimal fill vector* $\boldsymbol{q}^*$*.*
3. *No explicit budget constraints appear. At the optimum, each MM k spends exactly* B_k*:* $\sum_{i\in \mathrm{MM}_k} c_i(p^*) q_i^* = B_k$*.*
4. *Clearing prices* $\boldsymbol{p}^*$ *are unique.*
5. *The program is solvable in polynomial time by any standard convex optimizer (interior point, projected gradient, etc.).*

Proof.

(1) Strict concavity. The objective is a sum of three terms:

- $B_k \ln\left(\sum_{i\in \mathrm{MM}_k} w_i q_i\right)$: the composition of ln (concave, increasing) with a linear function (concave). By the composition rule, this is concave. It is *strictly* concave because ln is strictly concave: for $\boldsymbol{q}^1 \neq \boldsymbol{q}^2$ with $U_k(\boldsymbol{q}^1) \neq U_k(\boldsymbol{q}^2)$, the strict inequality of ln applies.
- $\sum_{j\notin \text{ MM}} w_j q_j$: linear, hence concave.
- $-C_b(\boldsymbol{D}(\boldsymbol{q}))$: C_b is convex (log-sum-exp) and $\boldsymbol{D}$ is linear in $\boldsymbol{q}$, so $C_b(\boldsymbol{D}(\boldsymbol{q}))$ is convex, hence $-C_b$ is concave.

The sum of concave functions is concave. It is strictly concave because the $B_k \ln U_k$ term is strictly concave whenever $B_k > 0$ and MM k has at least two linearly independent order directions (so U_k is non-constant on $\mathcal{C}$).

(2) Uniqueness. Strict concavity on a compact convex set ($\mathcal{C}$ is a polytope) gives a unique maximizer.

(3) Budget absorption. The KKT conditions for MM order i belonging to agent k:

$$\frac{\partial}{\partial q_i}[B_k \ln U_k - C_b] = \frac{B_k w_i}{U_k} - p_{m(i)} - \lambda_i^+ + \lambda_i^- = 0$$

where λ_i^+, λ_i^- are box constraint multipliers and $p_{m(i)} = \partial C_b / \partial D_{m(i)}$ is the clearing price. For interior fills ($0 < q_i < \overline{Q}_i$):

$$\frac{B_k w_i}{U_k} = p_{m(i)} = c_i(\boldsymbol{p})$$

The *marginal rate of substitution* equals the clearing price. Multiplying by q_i and summing over $i \in \mathrm{MM}_k$:

$$\sum_{i\in \mathrm{MM}_k} \frac{B_k w_i q_i}{U_k} = \sum_{i\in \mathrm{MM}_k} c_i(\boldsymbol{p}) q_i$$

The left side simplifies: $\frac{B_k}{U_k} \cdot \sum w_i q_i = \frac{B_k}{U_k} \cdot U_k = B_k$. Therefore:

$$\sum_{i\in \mathrm{MM}_k} c_i(\boldsymbol{p}^*) q_i^* = B_k$$

Each MM spends its *entire* budget at the optimum. The budget constraint is not imposed — it *emerges* from the $B_k \ln U_k$ objective. This is the Eisenberg-Gale mechanism: logarithmic utility + budget weight B_k = implicit budget constraint.

(4) Price uniqueness. Prices are $p_k = \partial C_b / \partial D_k = \mathrm{softmax}(\boldsymbol{D}/b)$, a continuous function of the unique $\boldsymbol{q}^*$.

(5) Polynomial solvability. The objective is concave and C^∞ (for $b > 0$, all components are smooth). The feasible set is a polytope. Standard interior-point methods solve this in polynomial time. □

The C_b interaction is trivial — and that's the point. In the risk-neutral model, the entropy smoothing (C_b) was the *only* source of concavity, creating the isospectral deadlock with the budget non-convexity (both $O(1/b)$). In the risk-averse model, the $B_k \ln U_k$ term provides b-independent curvature of order $O(B_k/U_k^2)$. The $-C_b$ term adds *more* concavity (it helps, not hurts). Even at $b = 0$ (pure LP minting cost), the program remains strictly concave:

$$P_0^{\mathrm{RA}} : \quad \max_{\boldsymbol{q} \in \mathcal{C}} \quad \sum_k B_k \ln U_k + \sum_{j \notin \mathrm{MM}} w_j q_j - \max_k D_k$$

The $-\max_k D_k$ term is concave (not strictly), but the ln term provides strict concavity. Uniqueness holds at every temperature, including $b = 0$. **Annealing is unnecessary.**

The Fisher market isomorphism. Program P_b^{RA} is structurally isomorphic to the Eisenberg-Gale convex program for Fisher market equilibrium:

Component	**Fisher market**	**Risk-averse batch auction**
Buyers	Consumers with budgets B_k	MMs with budgets B_k
Goods	Divisible commodities	Outcome shares
Supply	Fixed endowment s_j	Minting (endogenous, cost C_b)
Utility	$U_k = \sum w_i x_{ki}$	$U_k = \sum w_i q_i$
Objective	$\sum B_k \ln U_k$	$\sum B_k \ln U_k + \text{retail} - C_b$
Prices	Dual of supply constraint	Dual of balance constraint = softmax

The prediction market extends Fisher markets in two ways: (1) supply is endogenous (minting adds shares at cost C_b, not fixed), and (2) non-MM ("retail") orders contribute linear welfare alongside the log-utility MMs. Both extensions preserve concavity.

What changes economically. Under P_b^{RA}, MMs with larger budgets B_k have proportionally more influence on prices (through the B_k weight), but with diminishing marginal impact on any single market. An MM concentrating capital on one market faces the ln penalty: the first dollar of fill generates infinite marginal utility, the last dollar generates vanishing utility. This naturally diversifies MM capital across markets — the ln enforces the flash-liquidity pattern (spread capital thin) without protocol-level constraints.

What doesn't change. Non-MM orders are still matched by UCP at clearing prices. The minting mechanism is unchanged. Prices are still softmax (for $b > 0$) or LP duals (for $b = 0$). The only difference is in how MM fills are *sized* relative to each other: proportional to welfare-per-dollar rather than absolute welfare.

8.8.15. Landscape of uniqueness results

Result	Condition	What's unique	Strength
Proposition 6	No budgets	Prices (Fenchel dual)	Unconditional

Theorem 8	DIC holds at b	Fill vector + prices	Checkable certificate
Contraction	$b > \|A\|_\infty /2$	Everything (exponential)	Checkable, global
Proposition 5	$\boldsymbol{\mu} = 0$	Demands + prices	Unconditional (slack budgets)
Theorem 9	Generic θ	Everything	Almost everywhere
Theorem 10	Kelly utility ($B_k \ln U_k$)	Everything	Unconditional (modified model)
Proposition 7	—	—	Impossible (isospectral)

8.8.16. Connection to related frameworks

Quantal Response Equilibria. The Diluted Influence Condition is the prediction-market analog of the contraction condition for Logit QRE (McKelvey & Palfrey, 1995). In a QRE, agents choose actions via softmax of expected payoffs; uniqueness holds when the temperature is high enough relative to the payoff sensitivity. Our condition $Q \leq 2b/(2p-1)$ is the market-specific contraction bound. Theorem 9 extends this: just as QRE uniqueness holds for generic payoff matrices even when the temperature is low (McKelvey & Palfrey, Theorem 5), our generic uniqueness holds for all $b > 0$.

Negative semidefinite games. Hofbauer and Sandholm (2007) proved uniqueness of logit equilibria for *negative semidefinite* games — games where increasing adoption of a strategy decreases its payoff. Our market is exactly this: buying more of outcome k raises p_k, reducing the marginal payoff to further buyers. Their result covers the unconstrained case; our Theorem 8 extends it to budget-constrained agents under DIC, and Theorem 9 extends it to budget-constrained agents for generic parameters.

Budget additivity. Devanur and Dudík (2015) showed that budget-constrained LMSR exhibits *budget additivity*: two agents with budgets B_1, B_2 and identical beliefs affect prices identically to one agent with budget $B_1 + B_2$. This is a manifestation of hidden convexity — the bilinear budget boundaries merge into a convex hull in the dual space. Theorem 10 recovers this convexity for batch auctions by adopting the same economic model (diminishing-returns agents): the Eisenberg-Gale program is explicitly convex. Whether the *risk-neutral* batch auction also has hidden convexity (which would strengthen Theorem 9 from generic to unconditional) remains open — Proposition 7 shows it cannot come from the standard Fenchel dual alone.

Parametric optimization. Theorem 9 uses the Parametric Transversality Theorem from differential topology (Abraham & Robbin, 1967; Guillemin & Pollack, 1974). The technique is standard in economic theory: Debreu (1970) used it to prove generic finiteness of Walrasian equilibria, and Mas-Colell (1985) applied it to smooth economies. Our application is new in the prediction market context.

9. Discussion

9.1. What the Proof Establishes

The central result is that LP batch auction clearing and LMSR pricing are the *same mathematical object at different temperatures*:

Property	LP ($b = 0$)	LMSR ($b > 0$)
Minting cost	$\max_k D_k$	$b \ln \sum \exp(D_k/b)$
Price constraint	$\boldsymbol{p} \in \Delta$ (hard)	$-bH(\boldsymbol{p})$ penalty (soft)

Price function	Set by marginal order	Softmax of demand
Liquidity	Endogenous (order book)	Exogenous (parameter b)
Market maker loss	Zero	$\leq b \ln K$
Scaling (mutual exclusion)	$O(K)$	$O(K)$ smoothed / $O(2^K)$ combinatorial

9.2. Convergence of Optimizers

Theorems 1–3 establish the structural connection between LP clearing and LMSR. The following two results complete the convergence picture.

Theorem 5 (Optimizer Convergence). *Let $\boldsymbol{q}_b^*$ denote an optimal fill vector of P_b for each $b > 0$, and let $\mathcal{Q}^*$ denote the set of optimal fill vectors of P (the LP). Then:*

$$\lim_{b \to 0^+} \operatorname{dist}(\boldsymbol{q}_b^*, \mathcal{Q}^*) = 0$$

If P has a unique optimum $\boldsymbol{q}^*$, then $\boldsymbol{q}_b^* \to \boldsymbol{q}^*$.

Proof sketch. The feasible set $\left[0, \overline{\boldsymbol{Q}}\right]$ is compact. By Proposition 1, the objectives $f_b(\boldsymbol{q}) = \sum w_i q_i - C_b(\boldsymbol{D}(\boldsymbol{q}))$ converge uniformly to $f(\boldsymbol{q}) = \sum w_i q_i - V(\boldsymbol{D}(\boldsymbol{q}))$ on this compact set. The result follows from Berge's Maximum Theorem: the argmax correspondence of a uniformly convergent sequence of continuous functions on a compact set is upper hemicontinuous. □

Theorem 6 (Exponential Price Convergence). *Let $\boldsymbol{p}(b)$ denote the LMSR prices at temperature b, and let $k^* = \operatorname{argmax}_k D_k$ with gap $\Delta = D_{k^*} - \max_{k \neq k^*} D_k > 0$. Then:*

$$|p_{k^*}(b) - 1| \leq (K - 1) \cdot \exp(-\Delta / b)$$

$$p_k(b) \leq \exp(-\Delta / b) \quad \text{for } k \neq k^*$$

The prices converge exponentially fast in $1/b$, with rate governed by the demand gap Δ.

Proof sketch. Divide numerator and denominator of the softmax by $\exp(D_{k^*}/b)$:

$$p_k(b) = \frac{\exp((D_k - D_{k^*})/b)}{1 + \sum_{j \neq k^*} \exp\left(\left(D_j - D_{k^*}\right)/b\right)}$$

For $k \neq k^*$: the numerator is $\leq \exp(-\Delta/b)$ and the denominator is ≥ 1. For $k = k^*$: $1 - p_{k^*} = \sum_{k \neq k^*} p_k \leq (K - 1) \exp(-\Delta/b)$. □

Remark. In the degenerate case $\Delta = 0$ (tied demands), the softmax splits mass equally among tied outcomes. The LP can choose any split, so convergence is to the *set* of LP optima (Theorem 5), not to a unique point. For generic instances (almost all order books), $\Delta > 0$.

9.3. Open Problems

1. **Extension to non-exclusive groups.** When markets are correlated but not mutually exclusive, group minting doesn't apply directly. The marginal decomposition (paper §4) handles this approximately; the exact connection to combinatorial LMSR in this regime is open.

2. **Risk-averse clearing in practice.** Theorem 10 proves that the Eisenberg-Gale program P_b^{RA} is convex and has a unique optimum. The open question is *quantitative comparison*: how much welfare does the risk-averse model sacrifice relative to the risk-neutral LP? When $B_k \gg U_k$ (large budgets, small fills), the ln objective approaches linear welfare and the gap should vanish. Formalizing this convergence — $P_b^{\mathrm{RA}} \to P_b$

as $B_k \to \infty$ with fixed welfare — would establish risk-averse clearing as a practical drop-in replacement.

9.4. Connection to Prior Work

Abernethy, Chen, and Vaughan (2013): Proved that any cost-function market maker satisfying a set of axioms must price via a convex cost function, with LMSR corresponding to the entropy conjugate. Our Theorem 2 is the batch-auction analog of their result: the LP minting cost is the "simplest" (piecewise linear) cost function satisfying the axioms, with LMSR as its smooth relaxation.

Fortnow, Kilian, Pennock, and Wellman (2005): Used LP for combinatorial call market matching. Our group minting variable provides the structural reason *why* LP works: it directly encodes the $O(K)$ mutual exclusivity constraint that combinatorial approaches must discover.

Agrawal, Wang, and Ye (2011): Proposed convex pari-mutuel call auction mechanisms using convex programming. Our framework is a specialization where the convex cost function has a specific form ($V = \max$) motivated by prediction market minting.

Chen and Pennock (2007): Utility framework for bounded-loss market makers. The parameter b in our framework corresponds directly to their loss bound. Our contribution is showing that $b = 0$ (zero loss) is achievable in a batch auction setting.

Devanur and Dudík (2015): Proved price uniqueness for budget-constrained sequential LMSR via Bregman divergence, and budget additivity. Our Theorem 8 proves primal uniqueness under DIC; Theorem 9 proves generic uniqueness for the risk-neutral model; Theorem 10 achieves unconditional uniqueness for the risk-averse model via the same Eisenberg-Gale structure underlying their budget additivity result.

Hofbauer and Sandholm (2007): Proved uniqueness of logit equilibria for negative semidefinite games. Our prediction market is negative semidefinite (price rises with demand). Their result covers the unconstrained case; our Theorems 8–9 extend to budget-constrained agents.

McKelvey and Palfrey (1995): Established the Quantal Response Equilibrium framework. The Diluted Influence Condition is the prediction-market analog of their high-temperature uniqueness result; Theorem 9 extends to generic uniqueness at all temperatures.

Rockafellar (2023): Augmented Lagrangian framework for variational convexity. Provides local uniqueness (tilt stability) at non-degenerate KKT points — the mechanism underlying Step 2 of Theorem 9′s homotopy argument.

Abraham and Robbin (1967); Debreu (1970): The Parametric Transversality Theorem used in Theorem 9 is standard in economic theory: Debreu used it to prove generic finiteness of Walrasian equilibria. Our application to prediction market clearing with budget constraints is new.

Eisenberg and Gale (1959): Convex program for Fisher market equilibrium via expenditure variables. Our Theorem 10 establishes that prediction markets with risk-averse MMs ($B_k \ln U_k$ welfare) are isomorphic to Fisher markets: the Eisenberg-Gale program with endogenous supply (minting cost C_b) is convex. Proposition 7 identifies why the same approach fails for risk-neutral MMs: constant marginal returns provide no curvature to counteract the $O(1/b)$ budget non-convexity.

Next steps: (1) Empirical validation: implement LP solver with annealing, compare to MILP baseline. (2) Implement risk-averse clearing (Theorem 10) as a convex program — no annealing needed. (3) Quantitative comparison: welfare gap between P_b^{RA} *and* P_b *across realistic order books.*